Reviewer Pack — Kronecker Coefficient Gap in Symmetric Decompositions of Per…

Entry b1e1fdcf3c47 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-12 11:18:21 UTC

Kronecker Coefficient Gap in Symmetric Decompositions of Permanent Polynomials

Entry ID: `b1e1fdcf3c47` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-12 11:18:21 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory of Symmetric Groups (Kronecker coefficients) **Field B** (complexity object): Boolean Circuit Size for Permanent Polynomial

Statement:

> For a random n×n matrix M with entries ±1, the Kronecker coefficient g(λ, μ, ν) for the decomposition of Sym^2(Perm_n) ⊗ Sym^2(Perm_n) is Ω(n^2) and O(n^3), where λ, μ, ν are partitions of 2n. The ratio g(λ, μ, ν)/n^2 is strictly greater than 1 for all n ≤ 40.

Rationale (proposer's reasoning):

> Kronecker coefficients encode the multiplicity of irreducible representations in tensor products, which could reflect the inherent symmetry complexity of permanent polynomials. Their growth rate may correlate with the minimal circuit size required to compute the permanent, as symmetric decompositions often require more terms than antisymmetric ones.

Taxonomy category: `GCT_DET_PERM` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `64f4861003016fb3`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	SAFE	0.95	SAFE	UNCERTAIN

Barrier	Verdict	Confidence	LLM₁	LLM₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	SAFE	SAFE
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.2s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def sign(x):
        return 1 if x >= 0 else -1

    def binomial(n, k):
        if k > n:
            return 0
        result = 1
        for i in range(k):
            result *= (n - i)
            result //= (i + 1)
        return result

    def kronecker_coefficient(λ, μ, ν):
        if len(λ) != len(μ) or len(μ) != len(ν):
            return 0
        n = len(λ)
        coeff = Fraction(1)
        for i in range(n):
            coeff *= binomial(λ[i] + μ[i], λ[i]) * binomial(μ[i] + ν[i], μ[i])
            coeff //= (binomial(λ[i] + ν[i], λ[i]) * binomial(ν[i] + λ[i], ν[i]))
        return abs(coeff)

    def perm(n):
        if n == 0:
            return [[]]
        result = []
        for i in range(n):
            for p in perm(n - 1):
                if not any(p[j] == i for j in range(len(p))):
                    result.append([i] + p)
        return result
```

```python
def sym_power(poly, k):
    result = [poly[0]]
    for _ in range(1, k):
        new_poly = [0] * (len(result) + len(poly) - 1)
        for i in range(len(result)):
            for j in range(len(poly)):
                new_poly[i + j] += result[i] * poly[j]
        result = new_poly
    return result

def tensor_product(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
```

```
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_antisymmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
```

```
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric(poly1, poly2):
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric(poly1, pol
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric(
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric(
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
```

```python
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
    result = [0] * (len(poly1) * len(poly2))
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            if i < j:
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            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

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            if i <= j:
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            if i < j:
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            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
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                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
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            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

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            else:
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def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
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    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_
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            if i <= j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
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def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
        for j in range(len(poly2)):
            if i < j:
                result[i * len(poly2) + j] = poly1[i] * poly2[j]
            else:
                result[i * len(poly2) + j] = -poly1[i] * poly2[j]
    return result

def kronecker_product_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_
    result = [0] * (len(poly1) * len(poly2))
    for i in range(len(poly1)):
```

```
            for j in range(len(poly2)):
                if i <= j:
                    result[i * len(poly2) + j] = poly1[i] * poly2[j]
                else:
                    result[i * len(poly2) + j] = -poly1[i] * poly2[j]
        return result

    def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
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                else:
                    result[i * len(poly2) + j] = -poly1[i] * poly2[j]
        return result

    def kronecker_product_antisymmetric_symmetric_antisymmetric_symmetric_antisymmetric_symmetric_
        result = [0] * (len(poly1) * len(poly2))
        for i in range(len(poly1)):
            for j in range(len(poly2)):
                if i < j:
# ... [truncated, full source in replay tarball]
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c507cd2f.py", line 97, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c507cd2f.py", line 81, in run_trial
    ratio = coeff / (n**2)
            ~~~~~~^~~~~~~~
OverflowError: integer division result too large for a float
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> Test crashed with overflow error during ratio calculation, preventing verification of conjecture for n ≤ 40 | next: Implement arbitrary-precision arithmetic or reformulate ratio calculation to avoid overflow

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	76647
2	preregistration	ollama_remote	qwen3:8b	0	0	24541
3	novelty	ollama_remote	qwen3:8b	0	0	20873
4	novelty	ollama_remote	qwen3:8b	0	0	14835
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13360
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	84039
7	judge	ollama_remote	qwen3:8b	0	0	18507

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 252802 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/b1e1fdcf3c47.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/b1e1fdcf3c47.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/b1e1fdcf3c47.tar.gz` (if generated)